Answer **one** question from this section in the spaces provided.

8 A hot air balloon begins to ascend vertically from level ground at time $t = 0$ s.

Figure 8.1 shows the variation with t of the velocity v of the hot air balloon during its ascent from $t = 0$ to $t = 25$ s.

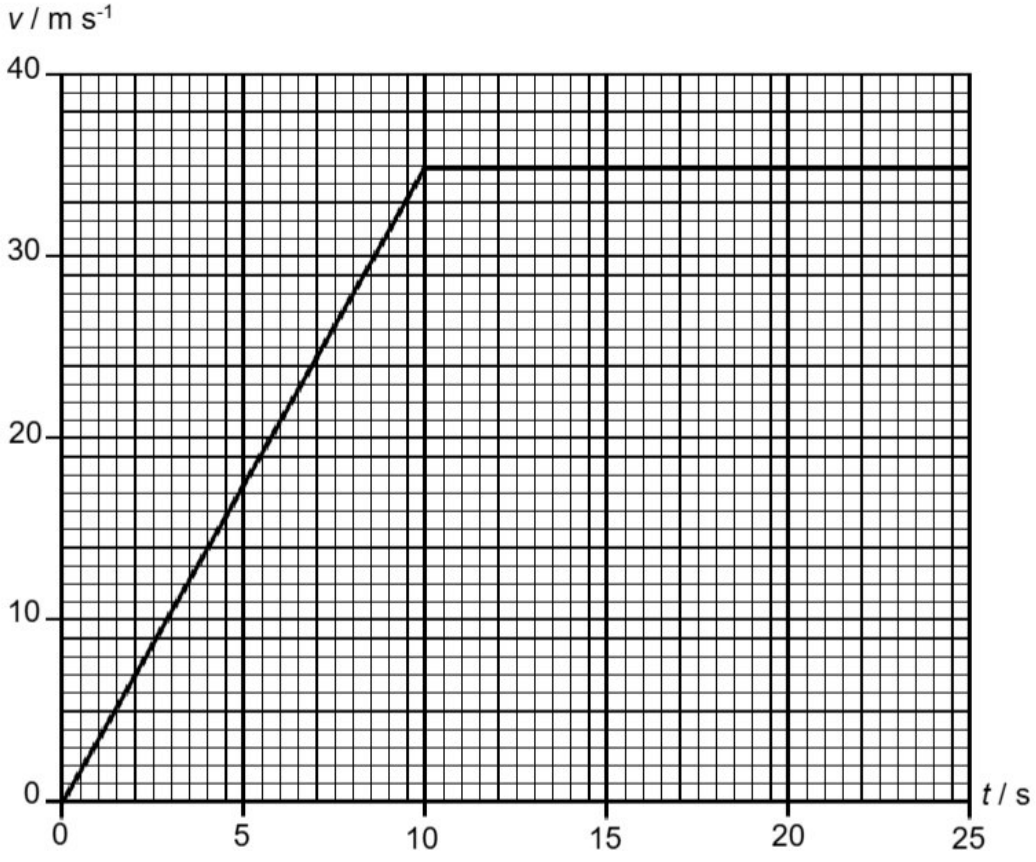


Figure 8.1

(a) Using Figure 8.1, calculate the acceleration of the hot air balloon from $t = 0$ to $t = 10$ s.

acceleration = ... m s^{-2} [1]

(b) At $t = 3.0$ s, a boy in the basket of the balloon realises that he has left his phone on the ground.

His friend, who is standing on the ground, throws the phone vertically upwards at this instant with a speed of 30 m s^{-1}.

The speed is insufficient for the phone to reach the boy.

(i) On Figure 8.1, sketch the variation with t of the velocity of the phone until the velocity is zero. [1]

(ii) Hence or otherwise, estimate the minimum separation between the boy and the phone.

separation = ... m [3]

(c) At t = 15.0 s, the boy drops a ball vertically downwards from the basket of the balloon.

Air resistance is negligible.

(i) Calculate the time taken for the ball to reach the ground after it is dropped.

time = ... s [2]

(ii) On Figure 8.2, sketch the variation with time t of the vertical displacement s of the ball from the instant it is released until it hits the ground.

Label this graph A. Include known values on the graph. [2]



Figure 8.2

(iii) At the same instant t = 15.0 s, the boy also releases a plastic ball for which air resistance is not negligible.

On Figure 8.2, sketch the variation with time t of the vertical displacement s of the plastic ball.

Label this graph B.
Assume that the plastic ball reaches terminal speed before hitting the ground.

[3]

(d) Another ball of mass 500 g, thrown by the boy, hits the ground at a speed of 11.5 m s^{-1} at an angle of 50° to the horizontal, as shown in Figure 8.3.

The ball rebounds with speed v at an angle of 35° to the horizontal.

Assume that the ground is frictionless.

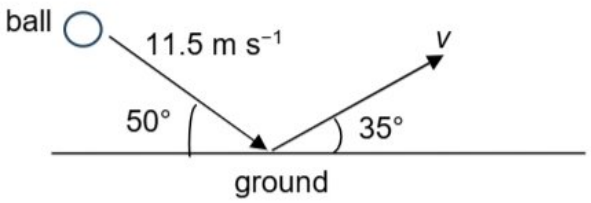


Figure 8.3

(i) Show that the rebound speed v of the ball is 9.0 m s^{-1}.

[2]

(ii) Calculate the magnitude of the change in velocity of the ball.

change in velocity = ... m s^{-1} [2]

(iii) The ball is in contact with the ground for a duration of 0.20 s.

Determine the magnitude of the average force exerted on the ball by the ground.

force = ... N [2]

(iv) State and explain how the rebound speed would change if friction between the ball and the ground were not negligible.

.....

.....

..... [2]

[Total: 20]